UNITED SERVICES
A KELSO INDUSTRIES COMPANY

TOOLBOX TALK 11 OF 52

Working with Compressed Gases and Cylinders

Trade: HVAC **Week:** 11 of 52 **Scheduled Date:** Monday, November 02, 2026 **Duration:** 10-15 min

Why This Matters

HVAC technicians handle oxygen, acetylene, nitrogen, and refrigerant cylinders daily for brazing, pressure testing, and system charging. A cylinder is a pressurized projectile if its valve is damaged - improper handling has caused fatal injuries when a cylinder became a missile after its valve sheared off.

Key Hazards

- Cylinder becoming a projectile if the valve is knocked off or damaged
- Explosion or fire from mixing incompatible gases or using damaged regulators
- Over-pressurizing a system with nitrogen during leak testing
- Oxygen enrichment near oil or grease causing violent ignition
- Falling or unsecured cylinders tipping over and causing crush injuries

Regulatory & Standards References

- OSHA 29 CFR 1926.350 - Gas Welding and Cutting (compressed gas handling)
- Compressed Gas Association (CGA) standards for cylinder handling and storage
- DOT regulations for cylinder transport

Best Practices / Toolbox Checklist

- Always secure cylinders in an upright position with a chain or strap, whether stored or in transport
- Keep valve protection caps on cylinders whenever they are not connected for use
- Never use oxygen or acetylene equipment interchangeably - use only gas-specific regulators and hoses
- Never use oil or grease on oxygen fittings or regulators - it can ignite violently
- Follow manufacturer-rated test pressures when pressure testing with nitrogen - never exceed system ratings
- Store oxygen and fuel gas cylinders at least 20 feet apart or separated by a fire-rated barrier
- Inspect regulators, hoses, and fittings for damage before each use
- Transport cylinders with valves closed and regulators removed

Discussion Questions

- How should cylinders be secured on our truck and at the work area?
- What is the maximum safe test pressure for the system we're pressure testing today?
- What would you do if you noticed a cylinder valve was damaged?

Crew Sign-In & Acknowledgment

By signing below, each employee acknowledges they participated in this toolbox talk and understand the hazards and best practices discussed.

#	Printed Name	Signature	Date
1			
2			
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Presented By: _____ **Supervisor / Foreman Sign-Off:** _____

Notes / Incidents / Follow-Up Items: _____